JUNE 2025

REGIONAL MARITIME UNIVERSITY
MARINE ENGINEERING DEPARTMENT
BME/BMT 27
FLUID MECHANICS I (BMAE 210)
END OF SEMESTER EXAMINATIONS

TIME ALLOWED: 3 HOURS

INSTRUCTIONS:
Answer ANY FOUR questions. All questions carry equal marks of 25.

Q1. (12 MARKS)

a) State the difference between the following types of fluid flow
   i. Steady and Unsteady flow
   ii. Uniform and Non-Uniform flow
   iii. Compressible and incompressible flow
   iv. Laminar and Turbulent flow

b) The weight of an object in air is 10 N. When it is submerged in a liquid of relative density 1.15, the volume of the liquid increased from 15cm³ to 20 cm³. What is the weight of the object in water? (5 MARKS)

c) A capillary tube 0.12 mm in diameter has its lower end immersed in liquid with density $760\ ^{kg}/_{m^3}$. Calculate the height of capillary rise if $\sigma = 0.054\ ^{N}/_{m}$, $\emptyset = 30°$, $g = 10\ ^{m}/_{s^2}$ (8 MARKS)

Q2.

a) State Bernoulli's Principle and the assumptions used in the derivation of the Bernoulli equation. (6 MARKS)

b) Discuss how Pascal's Law is applied in hydraulic systems. (4 MARKS)

c) Calculate the dynamic viscosity of an oil (in poise) which is used for lubrication between a square plate of size 60 cm by 60 cm, and an inclined plane with angle of inclination, 30°. The weight of the square plate is 250 N and it slides down the inclined plane with a uniform velocity of 0.3 m/s. The thickness of the oil film is 1.5mm. (10 MARKS)

d) d) A hydraulic press has a ram of 30 cm diameter and a plunger of 5 cm diameter. Find the weight lifted by the hydraulic press using pascals law, when a force of 400 N is applied at the plunger. (5 MARKS)

Q3.

a) a) Identify five (5) instances that cause minor energy losses in pipe systems. (5 MARKS)

b) A vessel's cooling water system is designed with a pipeline of 120 m in length and an internal diameter of 0.12 m, transporting seawater at a flow velocity of 3.5 L/s. The pipe is made of steel with a roughness of 0.025 mm. [$\rho_{seawater}$ = 1000kg/m³ and μ = 0.001 Pa.s]. Determine:
   i. the flow regime (laminar or turbulent).
   ii. the friction factor (f) using Churchill's equation
   iii. the head loss due to friction (H_f) with the Darcy-Weisbach equation. (20 MARKS)

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Q4.

a) a) State the difference between Newtonian and Non-Newtonian fluids and give two (2) examples of each. (6 MARKS)

b) b) Identify four (4) techniques for controlling cavitation in pumping systems. (4MARKS)

c) c) An open cylindrical vertical container is filled with water to a height of 30 cm above the bottom and over that an oil of specific gravity 0.82 for another 40 cm. Determine the absolute pressure at the bottom of the cylinder, if P_{atm} is 1 bar. (5 MARKS)

d) d) An oil of sp. gr. 0.8 is flowing through a venturimeter having inlet diameter 20 cm and throat diameter 10 cm. The oil-mercury differential manometer shows a reading of 25 cm. Calculate the discharge of oil through the horizontal venturimeter. Take Cd = 0.98. (10 MARKS)

Q5.

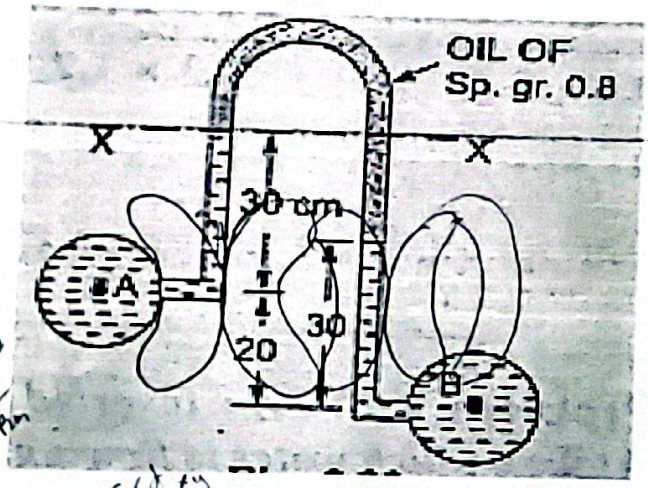
$Q = \frac{Cd\, a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}}$ $h = x$

a) How does the viscosity of a fluid vary with temperature? (4 MARKS)

b) Why do some people experience ear blockage or shortness of breath at high elevations? Briefly explain. (5 MARKS)

c) A jet of fluid with relative density 0.82 strikes a wall and is brought to rest from an initial velocity of 8 m/s. If the flow rate is 3.2 m³/s, determine the impact force. (6 MARKS)

d) In the figure shown below, an inverted differential manometer is connected to two pipes A and B which convey water. The fluid in the manometer is oil of specific gravity 0.8. For the manometer readings shown in the figure, find the pressure differential between points A and B. (10 MARKS)



Q6.

a) Distinguish between manometers and mechanical pressure gauges. What are the different types of mechanical pressure gauges that you know? (7 MARKS)

b) Why is the width (thickness) of a dam greater at the bottom than at the top? Explain your answer using the principles of fluid pressure. (5 MARKS)

c) A pump is used to transport water from a lower reservoir to an elevated tank 25 m above the pump. The flow rate through the pump is 40 L/s. The total head loss due to friction and fittings in the system is estimated to be 5 m. The pump consumes 15 kW of electrical power during operation. Determine

I. the total dynamic head (TDH) the pump must overcome.

II. the hydraulic power delivered to the water by the pump.

III. the efficiency of the pump. (13 MARKS)

INITIALS OF EXAMINER: B.D.

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